Marking Scheme
Mathematics
Class: X Maximum Marks: 80

General note: accept any answer that conveys the value points below, in the candidate's own words, unless the question demands a specific form. Award marks for the steps shown, not only for the final answer.

Section A – Objective Questions

Q1. (1 mark)
Answer: b) 6

Q2. (1 mark)
Answer: c) 5

Q3. (1 mark)
Answer: c) infinitely many solutions

Q4. (1 mark)
Answer: c) 5 and -2

Q5. (1 mark)
Answer: b) 29

Q6. (1 mark)
Answer: c) 10 units

Q7. (1 mark)
Answer: b) 1

Q8. (1 mark)
Answer: b) 1

Q9. (1 mark)
Answer: c) 1/2

Q10. (1 mark)
Answer: a) 3

Q11. (1 mark)
Answer: b) 154 cm^2

Q12. (1 mark)
Answer: c) 55

Q13. (1 mark)
Answer: c) 0

Q14. (1 mark)
Answer: b) -4

Q15. (1 mark)
Answer: a) 2 : 3

Q16. (1 mark)
Answer: b) 1/2

Q17. (1 mark)
Answer: c) 6

Q18. (1 mark)
Answer: c) 2

Q19. (1 mark)
Answer: c) 27 cm^3

Q20. (1 mark)
Answer: b) (4, 6)

Section B – Short Answers I

Q21. (3 marks)

- From x - y = 4, take x = y + 4 (or an equivalent first step)
- Substituting in x + y = 10 gives (y + 4) + y = 10, so y = 3
- Then x = 7; solution x = 7, y = 3

Mark split: 1 + 1 + 1

Q22. (3 marks)

- Sum of zeroes = 3 + (-2) = 1; product of zeroes = 3 x (-2) = -6
- The polynomial is x^2 - (sum of zeroes)x + (product of zeroes)
- So the polynomial is x^2 - x - 6 (any non-zero multiple is acceptable)

Mark split: 1 + 1 + 1

Q23. (3 marks)

- Here a = 5, d = 3, n = 20
- Using S = n/2 [2a + (n - 1)d] = 10 [10 + 19 x 3]
- Sum = 10 x 67 = 670

Mark split: 1 + 1 + 1

Q24. (3 marks)

- Write tan A = sin A / cos A, so 1 + tan^2 A = 1 + sin^2 A / cos^2 A
- Combining: = (cos^2 A + sin^2 A) / cos^2 A
- Using sin^2 A + cos^2 A = 1 gives 1 / cos^2 A = sec^2 A

Mark split: 1 + 1 + 1

Q25. (3 marks)

- Area of a sector = (angle / 360) x pi x r^2
- = (90 / 360) x (22 / 7) x 14 x 14
- = 154 cm^2

Mark split: 1 + 1 + 1

Q26. (3 marks)

- Arranging in ascending order: 11, 12, 14, 15, 16, 18, 20
- There are 7 observations, so the median is the 4th observation
- Median = 15

Mark split: 1 + 1 + 1

Section C – Short Answers II

Q27. (4 marks)

- First step: 3x^2 divided by x gives 3x; multiply back and subtract to get -x - 4
- Second step: -x divided by x gives -1; multiply back and subtract
- Quotient = 3x - 1
- Remainder = -2

Mark split: 1 + 1 + 1 + 1

Q28. (4 marks)

- Using the section formula with m : n = 2 : 1
- x = (2 x 8 + 1 x 2) / (2 + 1) = 18 / 3 = 6
- y = (2 x 3 + 1 x (-3)) / (2 + 1) = 3 / 3 = 1
- The required point is (6, 1)

Mark split: 1 + 1 + 1 + 1

Q29. (4 marks)

- By the Basic Proportionality Theorem, AD / DB = AE / EC
- Substituting: 3 / 6 = 4 / EC

- So EC = (6 x 4) / 3
- EC = 8 cm

Mark split: 1 + 1 + 1 + 1

Q30. (4 marks)

- Let h be the height of the tower; the tower, the ground, and the line of sight form a right-angled triangle
- tan 60 degrees = h / 30
- So h = 30 x tan 60 degrees = 30 x square root of 3
- Height of the tower = 30 x 1.732 = 51.96 m (approximately)

Mark split: 1 + 1 + 1 + 1

Q31. (4 marks)

- Total surface area of a cylinder = 2 x pi x r x (r + h)
- = 2 x (22 / 7) x 7 x (7 + 20)
- = 44 x 27
- = 1188 cm^2

Mark split: 1 + 1 + 1 + 1

Q32. (4 marks)

- Total number of possible outcomes = 6 x 6 = 36
- Favourable outcomes: (2, 6), (3, 5), (4, 4), (5, 3), (6, 2)
- Number of favourable outcomes = 5
- Required probability = 5 / 36

Mark split: 1 + 1 + 1 + 1

Section D – Long Answers

Q33. (6 marks)

- Let the breadth be b m; then the length is (2b + 1) m
- Area: b(2b + 1) = 528, giving 2b^2 + b - 528 = 0
- Discriminant = 1 + 4 x 2 x 528 = 4225 = 65^2, so b = (-1 + 65) / 4 = 16 (rejecting the negative root)
- Breadth = 16 m
- Length = 2 x 16 + 1 = 33 m (check: 16 x 33 = 528)

Mark split: 1 + 2 + 2 + 1

Q34. (6 marks)

- Figure and given: external point P, centre O, points of contact A and B; OA and OB joined, OP joined
- In triangles OAP and OBP: OA = OB (radii of the same circle)
- Angle OAP = angle OBP = 90 degrees (radius perpendicular to tangent)
- OP is common, so the triangles are congruent by RHS
- Therefore PA = PB by CPCT: the two tangents are equal in length

Mark split: 1 + 2 + 2 + 1

OR (alternative):

- Let the external point be P, the centre O, and the points of contact A and B
- In quadrilateral OAPB, the angles at A and B are each 90 degrees, since a radius is perpendicular to the tangent at the point of contact
- So the sum of the remaining angles: APB + AOB = 360 - 90 - 90
- Therefore angle APB + angle AOB = 180 degrees
- Hence the angle between the tangents is supplementary to the angle at the centre

Q35. (6 marks)

- Class marks of the classes are 12, 14, 16, 18, 20, 22, 24
- Products of frequency and class mark: 84, 84, 144, 234, 400, 110, 96
- Sum of frequencies = 64; sum of the products = 1152

- Mean = 1152 / 64
- Mean daily pocket allowance = 18 rupees

Mark split: 1 + 2 + 2 + 1